

FoodHub Order Data Analysis

Python Foundations Project

AI & ML Business Applications UT Austin – Great Learning

Presented by: Anoop Krishnan | 25 May 2026

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Executive Summary

● CONCLUSIONS (Q17)

- American (31%) & Japanese (25%) cuisines dominate - Top 3 cuisines account for 71% of all orders
- 71.2% of orders are on weekends (1,351 of 1,898) - FoodHub is primarily a weekend service
- Only 29.24% of orders exceed \$20, yet these generate ~60% of net commission revenue (\$3,689 of \$6,166)
- 38.78% of orders (736) are unrated -major customer-retention and feedback gap
- Cost, prep time and delivery time are uncorrelated (max $r = 0.04$) -each is an independent driver
- Weekday delivery averages 28.34 min vs 22.47 min on weekends -26% slower despite lower order volume
- 10.54% of orders (200) exceed 60 min total time (prep + delivery) -clear SLA risk

● RECOMMENDATIONS (Q17)

- Scale weekend staffing for American & Japanese restaurants to match demand without sacrificing speed
- Add post-order push-notification prompts to close the 38.78% unrated-order gap
- Introduce bundle / upsell deals to shift more orders above \$20 and boost high-commission revenue
- Feature and promote the 4 promo-eligible restaurants: Meatball Shop 4.51(*), Blue Ribbon FC 4.33(*), Shake Shack 4.28(*), Blue Ribbon Sushi 4.22(*) -all verified with >50 rated orders
- Fix weekday delivery lag -route optimisation & dynamic dispatch to bring avg below 25 min
- Implement 55-min total-time alert + priority dispatch to reduce 60+ min orders below 5%
- Expand loyalty rewards to the top 50 frequent customers; automate 20% vouchers at 8+ orders

Business Problem Overview and Solution Approach

- **PROBLEM STATEMENT**

- FoodHub is a food-aggregator app operating in New York City that connects customers to restaurants via a single smartphone platform. The company earns revenue through fixed commissions: 25% on orders >\$20, 15% on orders \$5-\$20, and 0% on orders <=\$5. As order volume grows, the data science team must analyse historical order data to understand demand patterns, improve customer experience, and optimise revenue.

- **SOLUTION APPROACH - 5-STEP EDA PIPELINE**

- Step 1 - Load Data : Read CSV (1,898 rows x 9 columns) into a pandas DataFrame
- Step 2 - Clean Data : Identify 'Not given' ratings as hidden NaN; verify no other missing values
- Step 3 - Univariate EDA : Explore distributions of all 9 variables individually (Q6-Q11)
- Step 4 - Multivariate : Analyse relationships & cross-variable patterns (Q12-Q16)
- Step 5 - Business Insights : Derive data-driven conclusions & actionable recommendations (Q17)

- **TOOLS USED**

- Python 3 | pandas & numpy | matplotlib & seaborn | Google Colab (Jupyter)

Data Overview

- **Q1 - DATASET SHAPE: 1,898 rows × 9 columns (each row = one unique food order)**

- **Q2 - DATA TYPES**

- int64 : order_id, customer_id, food_preparation_time, delivery_time
- float64 : cost_of_the_order
- object : restaurant_name, cuisine_type, day_of_the_week, rating

- **Q3 - MISSING VALUES**

- No standard NaN values in any column. However, the 'rating' column contains 736 entries labelled
- 'Not given' -a hidden missing value. These were replaced with NaN for correct numeric analysis.
- >> 736 of 1,898 orders (38.78%) have no customer rating.

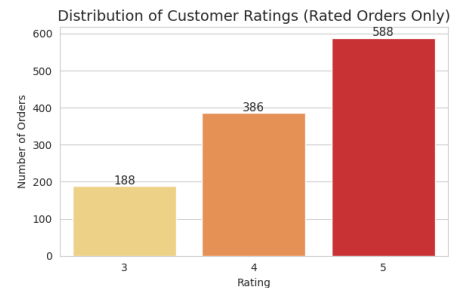
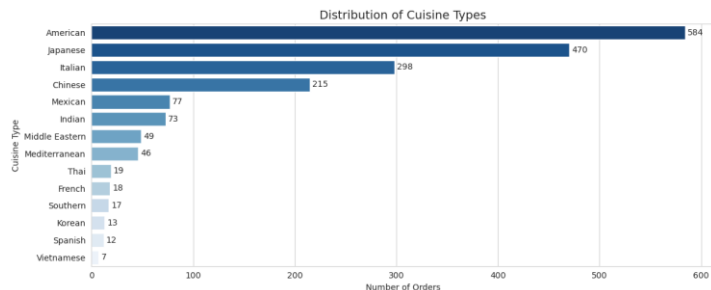
- **Q4 - DESCRIPTIVE STATISTICS**

- cost_of_the_order : Mean \$16.50 | Min \$4.47 | Max \$35.41 (right-skewed, median \$14.14)
- food_preparation_time: Mean 27.37 min | Range 20–35 min (near-uniform distribution)
- delivery_time : Mean 24.16 min | Range 15–33 min

- **Q5 - UNRATED ORDERS**

- 736 orders (38.78%) have rating = 'Not given'. Treating these as NaN ensures accurate
- mean/median calculations and avoids inflating or deflating the average customer rating.

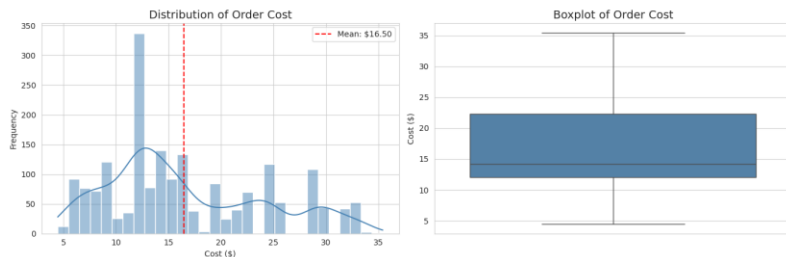
Univariate Analysis



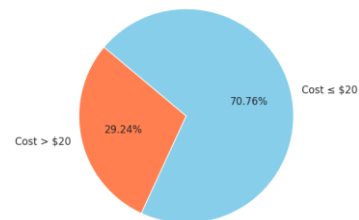
Q6 - Distribution of Cuisine Types

Q7 - Distribution of Customer Ratings

Order Cost Analysis



Percentage of Orders Costing More Than \$20



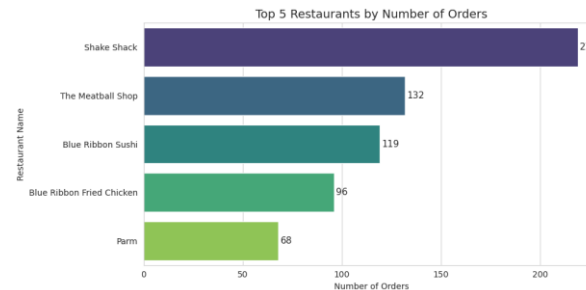
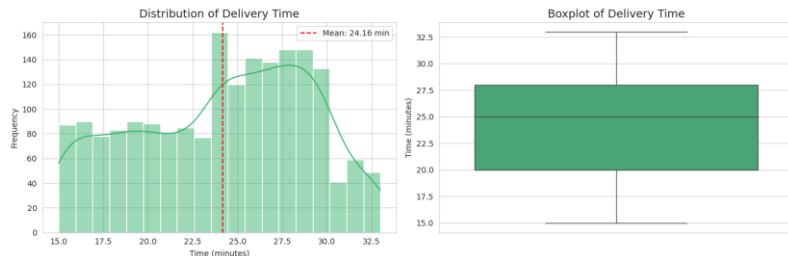
Q8 - Order Cost Distribution & Spread

Q9 - Orders Costing More Than \$20 (29.24%)

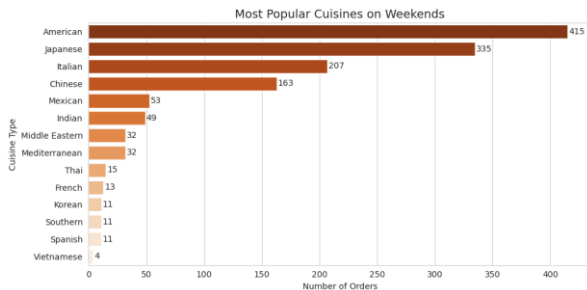
Key Findings: American (584) & Japanese (470) lead orders · Shake Shack #1 with 219 orders · 51.6% of rated orders = 5(*) · Only 29.24% of orders exceed \$20

Univariate Analysis (continued - Q8-Q11)

Delivery Time Analysis

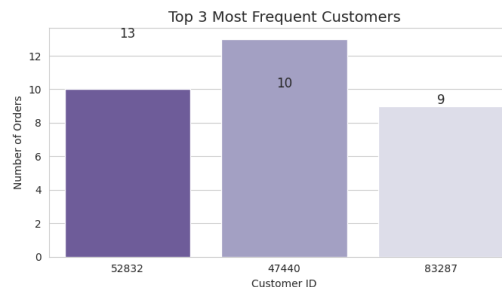


Q10 - Delivery Time Distribution



Q6 (Extra) - Most Popular Cuisines on Weekends

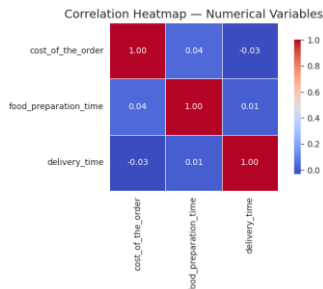
Q11 - Top 5 Restaurants by Order Count



Q11 (Extra) - Top 3 Most Frequent Customers

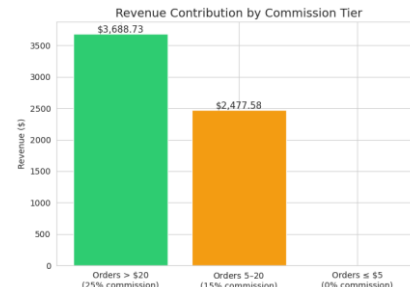
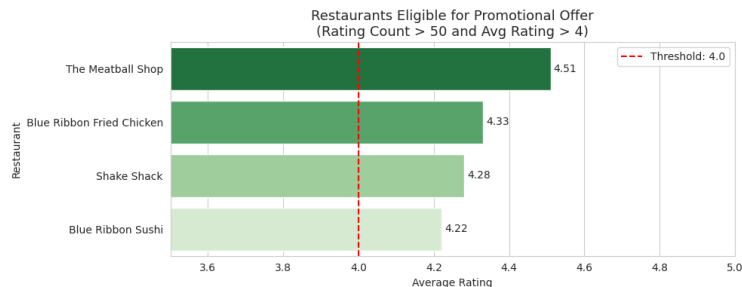
Key Findings: Weekend orders 2.5× weekday volume · American & Japanese peak on weekends · Mean cost \$16.50 · Mean delivery 24.16 min · Top customer #52832 placed 13 orders

Multivariate Analysis



Q12 - Correlation Heatmap: Cost, Prep & Delivery

Q12 - Order Cost: Weekday vs Weekend

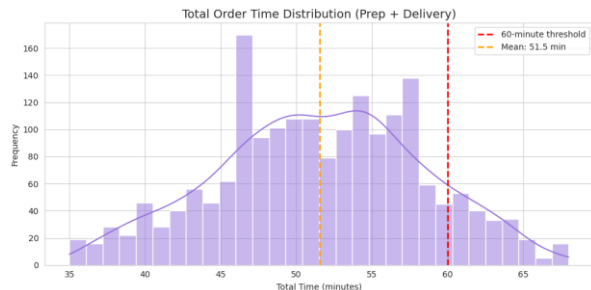


Q13 - Restaurants Eligible for Promotion

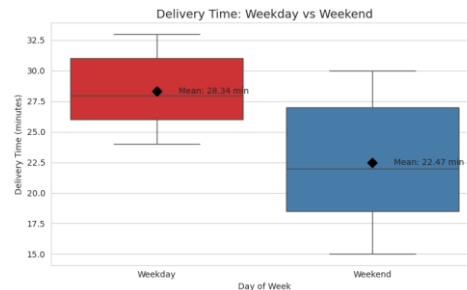
Q14 - Revenue by Commission Tier

Key Findings: Near-zero correlations between all 3 numeric variables · Weekday delivery 28.34 min vs 22.47 min weekend · 4 restaurants qualify for promotion · Net revenue = \$6,166.30

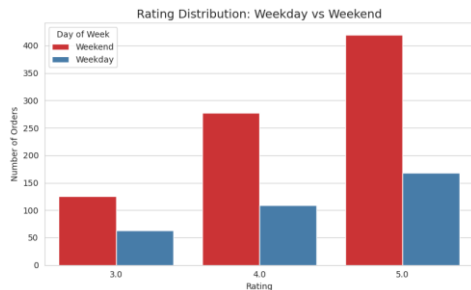
Multivariate Analysis (continued - Q12, Q15-Q16)



Q15 - Total Order Time Distribution (Prep + Delivery)



Q16 - Delivery Time: Weekday vs Weekend



Q12 (Extra) - Rating Distribution: Weekday vs Weekend

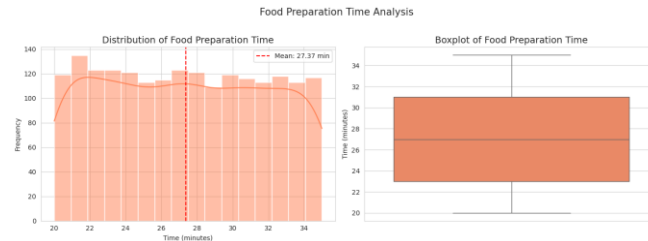


Q12 (Extra) - Prep Time vs Delivery Time (Scatter)

Key Findings: 10.54% of orders exceed 60 min total time · Mean total = 51.5 min · Weekend orders receive higher 5(*) ratings · No correlation between prep & delivery time

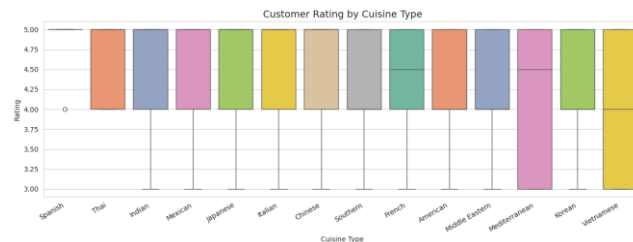
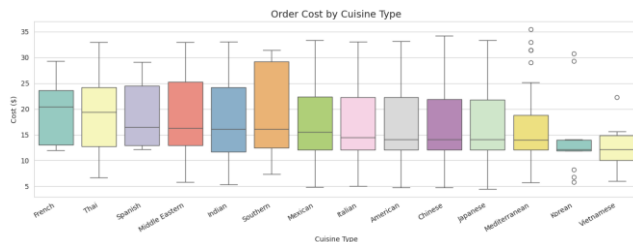
APPENDIX

Appendix - Supplementary Analysis & Data Reference



Q6 (Supp.) - Orders by Day of Week: Weekend vs Weekday

Q10 (Supp.) - Food Preparation Time Distribution



Q12 (Supp.) - Order Cost by Cuisine Type

Q12 (Supp.) - Customer Rating by Cuisine Type

Data Dictionary

order_id (int) | customer_id (int) | restaurant_name (str) | cuisine_type (str) | cost_of_the_order (float, USD) | day_of_the_week (str: Weekend/Weekday) | rating (str -> NaN if 'Not given')

Tools & Environment: Python 3 | pandas | numpy | matplotlib | seaborn | Google Colab | Dataset: 1,898 rows x 9 columns | Source: FoodHub order data



Happy Learning !

